

The Free-Boundary Conditions of RealQM, and their Overdetermination in Time

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Abstract

Real Quantum Mechanics (RealQM) represents N electrons by N unit charge densities carried on non-overlapping domains of ordinary three-dimensional space, the domains meeting at free boundaries whose location is not given in advance. In the established, real-valued theory — the one that computes atoms, molecules and nuclei — the ground state minimises the Coulomb energy, and on the free boundaries two conditions are to hold: continuity of the wave function (hence of the charge density) and a vanishing normal derivative (the homogeneous Neumann, or Bernoulli, condition). These two conditions fix the boundary and the static problem is well posed. We observe that the *time-dependent*, complex-valued extension — used to reach currents, magnetism and decay — changes this. A complex wave function carries a current, and requiring each electron to keep its unit charge as the boundary moves imposes a further condition, that the boundary move with the flow. Continuity, Neumann, and per-electron charge conservation are then three conditions on one free boundary: the time-dependent free-boundary problem is *overdetermined*. It cannot in general satisfy all three, and one must choose which to keep — keep the wave function continuous (the Bernoulli condition, with charge free to migrate between domains), or keep each electron's charge (with a jump in the wave function). We state the conditions plainly and record this fork; its resolution is left open. The real-valued static theory is unaffected: there the currents vanish, the boundary is at rest, and the two Bernoulli conditions hold without conflict.

1 The real-valued theory and its free boundary

RealQM represents N electrons by densities ψ_i^2 , each carried on its own region $\Omega_i \subset \mathbb{R}^3$, the regions non-overlapping, $\Omega_i \cap \Omega_j = \emptyset$ [1]. The wave functions ψ_i are real, each of unit charge,

$$\int_{\Omega_i} \psi_i^2 = 1, \tag{1}$$

and the ground state minimises the Coulomb energy

$$E = \sum_i \frac{1}{2} \int_{\Omega_i} |\nabla \psi_i|^2 + \frac{1}{2} \iint \frac{\rho(x)\rho(y)}{|x-y|} dx dy, \quad \rho = \sum_i q_i \psi_i^2, \tag{2}$$

over the wave functions and, jointly, over the partition into domains. Minimising over the ψ_i gives, on each domain, a one-electron eigenvalue equation in the self-consistent Coulomb field — so each

domain carries its own Poisson problem for its Hartree potential, coupled to the others only through the fields they produce.

The interfaces Γ between domains are not prescribed; they are located by the requirement that two conditions hold there.

(i) *Continuity*. The total $\Psi = \sum_i \psi_i$ is continuous, so at an interface between domains 1 and 2,

$$\psi_1 = \psi_2 \quad \text{on } \Gamma, \quad \text{hence} \quad \rho_1 = \rho_2. \quad (3)$$

Since $\rho_i = \psi_i^2$, continuity of the wave function *is* continuity of the charge density.

(ii) *Vanishing normal derivative* (homogeneous Neumann, the Bernoulli condition):

$$\frac{\partial \psi_i}{\partial n} = 0 \quad \text{on } \Gamma. \quad (4)$$

Condition (i) matches the two densities where the domains meet; condition (ii) fixes the location of the interface. With both imposed the static free-boundary problem is well posed, and it is this problem — unit charges, continuity, and Neumann, coupled by Coulomb — that reproduces the periodic table, chemical bonding, and, read for protons and electrons, nuclear binding [1, 2]. Everything here is static and real: the density is fixed in time, no charge flows across any interface, and the two conditions (3)–(4) sit together without tension.

2 The time-dependent extension carries a current

To reach phenomena that are absent from a static real density — currents, magnetism, radiation, decay — the wave functions are made *complex* and evolved by the Schrödinger equation generated by (2),

$$i \partial_t \psi_i = -\frac{1}{2} \nabla^2 \psi_i + q_i \phi[\rho] \psi_i, \quad (5)$$

with $\phi[\rho]$ the Coulomb potential of the total density. A complex ψ_i carries a current

$$\mathbf{j}_i = \text{Im}(\psi_i^* \nabla \psi_i), \quad \partial_t \rho_i + \nabla \cdot \mathbf{j}_i = 0, \quad (6)$$

which is identically zero when ψ_i is real. This current is what a static real density lacks and what the complex extension is for.

3 The free boundary is overdetermined in time

Now let the boundary move. Write V_n for its *normal velocity* — the speed at which the interface $\Gamma(t)$ advances along its unit normal $\mathbf{n}$ (for a level set $\Gamma = \{F = 0\}$, $V_n = -\partial_t F / |\nabla F|$; in one dimension, with the interface at $x = a(t)$, $V_n = \dot{a}$). It is a geometric quantity, the motion of the boundary itself, and is *a priori* distinct from the charge-flow velocity $\mathbf{j}_i / \rho_i$. The charge of electron i is $Q_i = \int_{\Omega_i(t)} \rho_i$, and with the interface in motion, using (6),

$$\frac{dQ_i}{dt} = \int_{\Gamma} (V_n \rho_i - \mathbf{j}_i \cdot \mathbf{n}) dS. \quad (7)$$

For each electron to keep its unit charge (1) as the boundary moves, the integrand must vanish, which is a *third* condition on the interface:

$$V_n = \frac{\mathbf{j}_i \cdot \mathbf{n}}{\rho_i} \quad \text{on } \Gamma, \quad (8)$$

the statement that the boundary moves with the charge flow. Imposed for both domains, (8) also demands $\mathbf{j}_1 \cdot \mathbf{n} / \rho_1 = \mathbf{j}_2 \cdot \mathbf{n} / \rho_2$, a matching of the flow speeds across the interface.

We now have *three* conditions to be satisfied on one free boundary: continuity (3), Neumann (4), and charge conservation (8). That is one condition too many. The free boundary carries a single location (one function on Γ) to be fixed, and the fields already obey their bulk equation (5); three interface conditions on that single freedom leave the time-dependent free-boundary problem *overdetermined*. It cannot, in general, satisfy all three at once.

Two remarks make the tension concrete. First, (4) and (8) clash directly: Neumann gives $\partial_n \psi_i = 0$, hence $\mathbf{j}_i \cdot \mathbf{n} = \text{Im}(\psi_i^* \partial_n \psi_i) = 0$, and then (8) forces $V_n = 0$ — the boundary cannot move at all. Second, continuity (3) is a condition on the *value* of ψ_i at the interface, whereas charge conservation (8) is a condition on its *gradient* (through $\mathbf{j}$); the two are independent, and holding both while the boundary moves is not automatic.

4 The fork

Faced with an overdetermined interface one must drop a condition. There are two natural choices.

- (A) *Keep the wave function continuous.* Retain continuity (3) and Neumann (4) — the same two conditions as the static theory — and let the boundary sit where they hold, moving as the fields evolve. The density stays continuous, Ψ remains regular, and the energy is conserved (with Neumann the interface contributes no energy flux and, by (3), no net energy from the moving boundary). What is given up is (8): as the boundary sweeps, charge migrates from one electron's domain into the other's, so the individual Q_i are not conserved (their sum is). This is the direct continuation of the static Bernoulli conditions, and it is the choice the existing RealQM level-set relaxation already realises.
- (B) *Keep each electron's charge.* Retain (8), so the boundary moves with the flow and every Q_i stays equal to one. The interface is then a contact discontinuity: the flow speeds match across it while the density is allowed to jump, so ψ (and ρ) is discontinuous and continuity (3) is given up. This conserves per-electron charge at the price of a wave function that is no longer continuous across a moving interface.

Both reduce to the static Bernoulli conditions at rest: when the currents vanish, (8) reads $V_n = 0$, the boundary stops, and continuity plus Neumann hold together. The two choices differ only when the boundary genuinely moves — that is, only in real-time, multi-domain dynamics.

5 Conclusion

The real-valued RealQM has a well-posed free boundary: unit charges, continuity of the wave function (hence of the density), and a vanishing normal derivative, coupled by Coulomb. These conditions sit together without conflict because the density is static and no charge flows. The

complex, time-dependent extension introduces a current, and with it a third interface condition — that each electron keep its charge as the boundary moves. Three conditions on one free boundary overdetermine it; the time-dependent free boundary cannot satisfy continuity, Neumann, and charge conservation together, and one must choose (A) a continuous wave function with charge free to migrate, or (B) conserved per-electron charge with a discontinuous wave function. Which the physics of real-time dynamics selects we leave open. It is worth being clear that this concerns only the time-dependent extension: the real-valued theory, which carries the program’s established results, is untouched — there the currents vanish, the boundary is at rest, and the Bernoulli conditions hold.

References

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